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E-202 NON SEGREGATED PHASE BUS DUCT

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of all non segregated phase bus duct as part of the work in accordance with Part 4, Subsection 1.2 Summary of Work.

The scope of work shall include, but not be limited to, all non segregated phase bus ducts from unit auxiliary transformer to unit MV switchgears and other connections such as PDC transformer, and LV switchgear connection (if required) as will be informed by the Owner during detailed design stage.

The above shall be complete including all accessories, hangers, supports, concrete inserts, minor steel as required, all hardware for assembly and connections and all other appurtenances required for a completely functional system.

Foundations for supports shall be furnished by the Contractor.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards.

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation - Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara

2.2 International Electrotechnical Commission

- IEC 60028 : International standard of resistance for copper
- IEC 60071 : Insulation co-ordination.
- IEC 60105 : Recommendation for commercial-purity aluminum busbar material.

IEC 60114 : Recommendation for heat-treated aluminum alloy busbar material of the aluminum - magnesium - silicon type.

IEC 60273 : Characteristic of indoor and outdoor post insulators for systems with nominal voltages greater than 1000 V.

2.3 American National Standards Institute

ANSI C29.1 : Test Methods for Electrical Power Insulators.

ANSI C37.20 : Standard for Switchgear Assemblies and Metal-Enclosed Bus.

2.4 American Society for Testing and Materials

ASTM A36/A36M : Standard Specification for Carbon Structural Steel.

ASTM A-90, A-123 : Galvanizing of Structural Steel Assemblies.

2.5 National Electrical Manufacturers Association

NEMA CC-1 : Electric Power Connectors for Substations

3. OPERATING AND DESIGN CONDITIONS

3.1 General

The non segregated phase bus duct shall be self-cooled, heavy duty, metal enclosed and designed for indoor and outdoor service. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental conditions at the Site.

3.2 Rating

Conductors, insulators and supports shall have the following ratings.

(1) 6.3 kV non segregated phase bus duct:

- Nominal voltage : 6.3 kV rms
- Maximum voltage : 7.2 kV rms
- Frequency : 50 Hz

- Continuous current : as required*)
- Momentary current : 125 kA peak*)
- Insulation level
 - a. Rated impulse withstand voltage : 60 kV
 - b. Rated power-frequency short duration withstand voltage (rms) : 20 kV

(2) 0.4 kV non segregated phase bus duct (if applicable)

- Nominal voltage : 0.4 kV rms
- Maximum voltage : 1.0 kV rms
- Frequency : 50 Hz
- Continuous current : as required *)
- Momentary current : 125 kA peak*)
- Insulation level
 - a. Rated impulse withstand voltage : 40 kV
 - b. Rated power-frequency short duration withstand voltage (rms) : 10 kV

Note: *) The Contractor shall be responsible for sizing of the conductor of non segregated phase bus duct. Calculation shall be submitted for approval of the Owner.

The hottest-spot temperature rise, at rated current, above an ambient temperature of 40°C of air surrounding the bus enclosures, shall not be more than:

- a. Bus conductors and their silver surfaced, welded, or brazed connections65°C.
- b. Bus enclosure and support structures where inaccessible.....70°C.
- c. Bus enclosure and support structures where easily accessible.....40°C.

The above temperature rating are in accordance with ANSI C37.20.

4. DETAILED REQUIREMENTS

4.1 General

Bus conductors and connections, insulators, supports, enclosures and supporting structures, when installed, shall have sufficient normal mechanical strength to ensure continuous and satisfactory operation under normal operating conditions and shall withstand without incurring damage the effect of any momentary current resulting from a three-phase, line-to-line or line-to-ground short circuit. The momentary current shall be as specified in Clause 3.2 above.

Joints between sections of the bus and between bus and switchgear terminals shall be with nuts, bolts and locking devices suitable for the bus duct materials concerned.

The Contractor shall design, supply and install all necessary support structures, which shall be hot-dipped galvanized steel and shall provide details of the foundations and all penetrations that are required. The support structure shall be grounded.

4.2 Bus Conductors, Insulators and Supports

The bus conductors shall be high conductivity copper. All contact surfaces for electrical connections shall be silver plated for copper conductor.

Bus conductors and bus supports shall be furnished with insulating coverings. Such insulation shall be tape or tube type, flame resistant, non-absorbent and oil and acid-resistant. All equipment and/or components shall be non-combustible and self-extinguishing.

Flexible connections shall be provided at all connections between bus and equipment terminals and between sections of the bus where required.

4.3 Bus Enclosures

Bus enclosures shall completely surround and enclose the bus conductors and insulators. They shall be fabricated, where necessary, of non-magnetic material and shall be designed to afford easy access to bus conductor joints and insulators for assembly, inspection and maintenance.

Bus enclosures shall be drip proof, ventilated type for indoor service and non-ventilated type for outdoor service. The bus shall have filter drains provided at equipment terminals, at building walls, floors and at all low points in the horizontal portions of the enclosures to ensure the elimination of any condensation that might collect and to prevent the accumulation of dirt within the enclosure during normal or abnormal operating conditions. Space heaters with thermostatic control shall be provided if needed to ensure satisfactory operation above the dew point at all times of the bus under the humid tropical conditions to which it will be subjected.

Housings shall be sectionalized in order to keep circulating current to a minimum.

4.4 Terminations and Connectors

Flexible laminated connectors shall be provided by the Contractor for bolted connection between the buses and transformers and switchgears. Flexible connectors shall be insulated to the same BIL as the bus conductors. Removable disconnecting links shall be provided.

4.5 Supporting Structures

The Contractor shall furnish all supporting structures, including columns, base plates, anchor bolts, grouting, hangers, hanger rods and necessary associated hardware required for hanging or supporting the indoor and outdoor sections of the bus. All steel structures shall be hot-dip galvanized. The support structure shall be grounded.

Insulating barriers shall be provided within the enclosures to seal points where the bus structure passes from indoor to outdoor to prevent condensation in outdoor section.

4.6 Ground Bus

The ground bus of different sections of non segregated phase bus shall be capable of carrying the rated momentary current for a period of at least two (2) seconds. The allowable time for carrying short circuit current shall be not less than that required to bring the bus voltage to zero including one (1) second operating time of the backup relays.

The ground bus shall be a continuous electrical path and may take the form of a separate copper bar installed within the non segregated phase bus or the bus enclosures itself may constitute the ground bus.

The ground bus dimensions shall be in no case be less than 7 x 50 mm copper or equivalent. All ground buses shall have provision for connections to the power plant grounding system.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 General

The Contractor may submit test certificates of tests which have been carried out on identical equipments. The type tests prescribed shall be carried out at the Contractor's expense in all cases where such certificates are not available or rejected.

5.2 Shop Tests

Tests shall be made to the non segregated phase bus duct in accordance with ANSI C37.20 or equivalent.

The following shop tests shall be performed by the Contractor.

- (1) High voltage tests.
- (2) Resistance measurements of bus conductors and joints.

5.3 Site Tests

The tests shall include, but not be limited to, the followings:

- (1) Insulation resistance test
- (2) High voltage tests.
- (3) Resistance measurements of bus conductors and joints.
- (4) Load tests to measure temperature rise and to locate any hot spots in the enclosure or supporting steel structures.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirement included in Part 4, Subsection 2.7.

The quality assurance/quality control procedures shall be as specified. These shall include, but not be limited to, the followings:

- (1) Welding
- (2) Nondestructive Testing
- (3) Packaging and Shipping
- (4) Shop Test Procedures
- (5) Cleaning and Painting

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Material certificates
- (2) NDE report
- (3) All shop test reports
- (4) All other documentation required by applicable codes and standards.

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